Gaussian Splatting

Code and reproduction instructions: `github.com/maxtlee/GaussianSplatting`. A from-scratch implementation: 2D Gaussians with covariance built from scale/rotation, a differentiable front-to-back compositing rasterizer, adaptive densification, and 3D Gaussians projected through a pinhole camera.

> **Reproducing.** `pip install -r requirements.txt` (Pascal GPUs: `requirements-cu126.txt`); `python scripts/check_env.py`; `python scripts/run_2d_suite.py --steps 2000`; `python scripts/run_3d.py --tag fixed --n 4000 --steps 1500`; `python scripts/run_batching.py`; `python scripts/make_figures.py`. All runs are seeded and resumable.

P1. 2D Gaussian and covariance from scale/rotation

A 2D Gaussian is a center μ, a covariance Σ, a color c, and an opacity α. Rather than optimize the four entries of Σ, it is built from a per-axis scale $S = \mathrm{diag}(s_x, s_y)$ and a rotation $R(\theta)$, $\Sigma = R\,S\,S^T\,R^T$, which is symmetric positive semi-definite by construction. The weight at a pixel is $w(x) = \exp(-\tfrac{1}{2}\,(x-\mu)^T\Sigma^{-1}(x-\mu))$ and the pixel alpha is opacity · w. The 2×2 inverse is computed analytically; the determinant is clamped away from zero for numerical safety. Unit tests check symmetry, PSD-ness (eigenvalues non-negative), axis-aligned correctness, and rotation invariance.

P2. Differentiable rasterizer (alpha compositing)

Gaussians are composited front to back, $C = \sum_i c_i\alpha_i T_i$, $T_i = \prod_{j<i}(1-\alpha_j)$, over a black background with straight RGB. Instead of a Python loop over Gaussians, the exclusive transmittance is the prefix product of $(1-\alpha)$ along the sorted axis (one `cumprod`) and the color sum is one matmul, so the whole composite is vectorized and differentiable. A test compares the vectorized result against an explicit per-Gaussian loop, and another checks a hand-computed two-Gaussian composite.

P3. Fit a 2D image

Each target is fit independently: Adam at 1e-2 for 2000 steps, scales in log space, colors and opacities through a sigmoid, from the prescribed spread-out initialization. The reported PSNR is $-10\log_{10}$ of the exact full-image MSE. Details and the quality-vs-count study are in P5.

P4. Densification (adaptive Gaussian count)

Score, split/clone, pruning. Every 200 steps each Gaussian is scored by the mean magnitude of the loss gradient with respect to its position accumulated since the previous pass, $g_i = (1/M)\sum_k \|\nabla_{\mu i} L_k\|_2$. I densify the top 25% by score (a fraction rather than the reference 2e-4 threshold, because the absolute gradient scale depends on the coordinate

system and a fixed threshold densified almost nothing at pixel scale). An under-fit Gaussian whose larger axis is at most 2% of the image width is **cloned** in place; otherwise it is **split** into two children offset ± half the major axis, with scales divided by 1.6. Gaussians with opacity below 0.005 are **pruned**, and the total is capped at the budget. Densification runs until 80% of the steps so the tail refines the grown set; the Adam moments of survivors are carried over and new Gaussians start from zero.

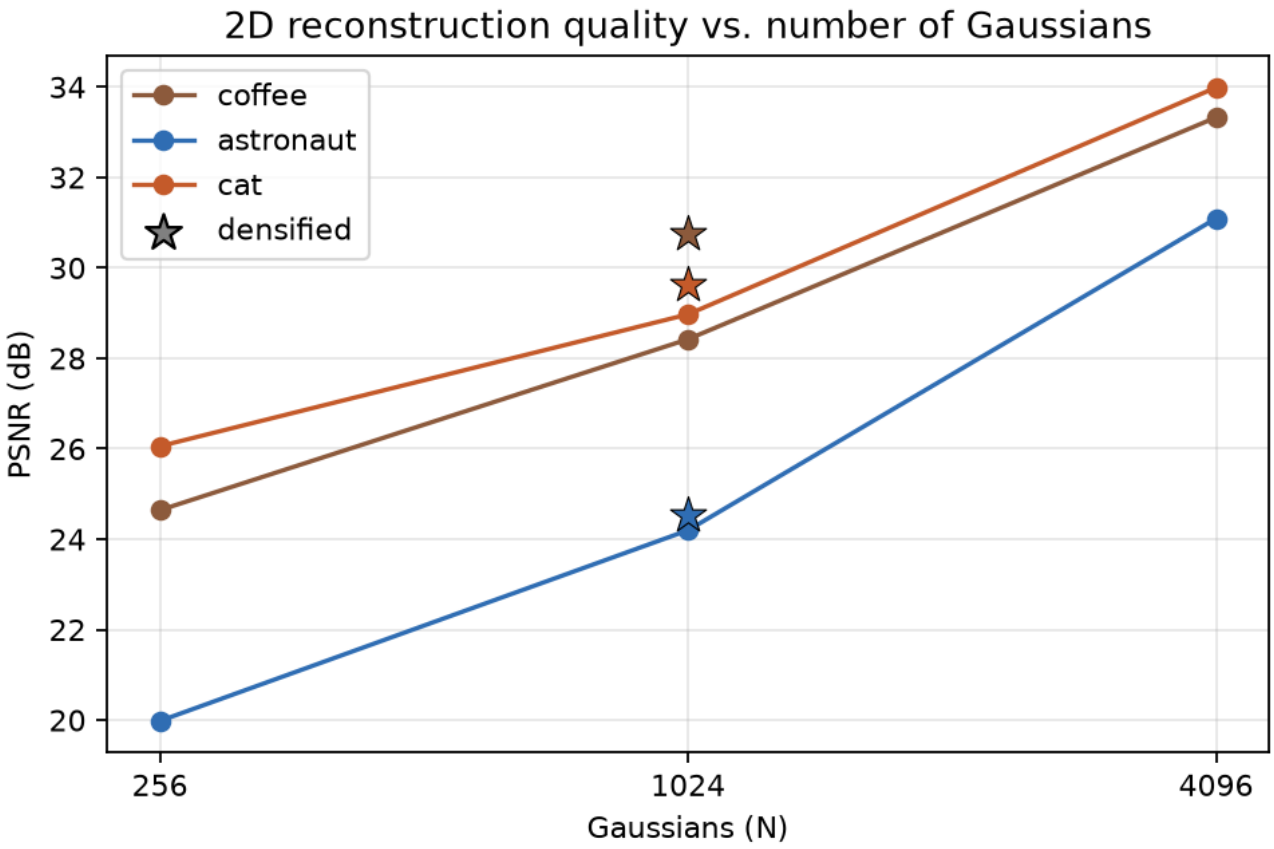
Result and head-to-head. On coffee, growing 256 → 1024 reaches **30.74 dB** versus **28.42 dB** for a plain fixed fit at the same 1024 Gaussians. Densification wins at equal count on all three targets (coffee +2.32, cat +0.65, astronaut +0.34 dB).



Densified coffee fit (target / render / 4x error), 30.74 dB at 1024 Gaussians.

P5. Results: quality vs. number of Gaussians

target	N = 256	N = 1024	N = 4096	densified (N = 1024)
coffee	24.64	28.42	33.33	30.74
astronaut	19.98	24.20	31.09	24.54
cat	26.06	28.97	33.99	29.62



PSNR against N, one curve per target (log x-axis); stars are the densified fits.

Which images need more Gaussians. astronaut needs the most: it is worst at N = 256 (19.98 dB) and gains 11.1 dB by N = 4096. It mixes smooth regions with sharp, high-contrast structure (helmet rim, white suit on a dark background, the flag), and sharp edges are what smooth blobs fit worst. coffee is dominated by large smooth regions, so it starts higher (24.64 dB) and still climbs to 33.33 dB as capacity captures the rim and reflections. cat is almost all high-frequency fur yet scores highest at low N (26.06 dB) and reaches 33.99 dB: dense texture has a small per-pixel amplitude around a dominant local color, so blobs match its local mean well even when the detail blurs — MSE tolerates that far more than a sharp edge. The lesson is that **structure (edges, contrast) matters alongside frequency content**: the image that needs the most Gaussians here is the one with the sharpest structure, not the one with the most high-frequency energy.

P6. 3D Gaussians and projection to a camera

The 3D covariance is built the same way, one dimension up, with a rotation from a unit quaternion (normalized first). Projection sends the mean through the pinhole camera and the covariance through the projection Jacobian at the mean: $\mu_{cam} = R_{wc}\mu + t$, $\Sigma' = J\,\Sigma_{cam}\,J^{T}$ with J the 2×3 derivative of the pixel projection, and depth = z_c. Gaussians behind the camera are given zero opacity. Tests check the quaternion-to-rotation matrix is orthonormal with determinant 1, that covariance is PSD and axis-aligned when the rotation is identity,

that a point on the optical axis projects to the principal point, and that an offset shrinks toward the center with depth.

P7. 3D scene reconstruction

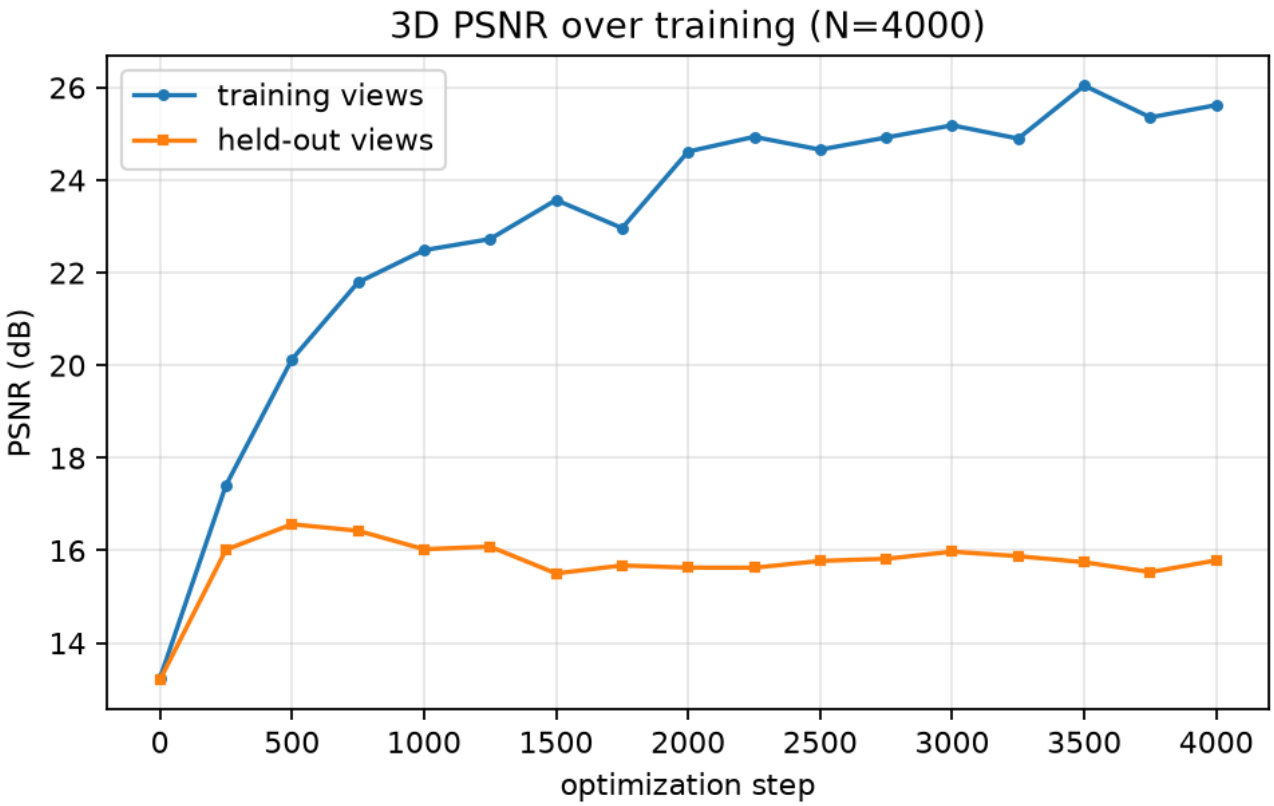
Each step samples one training camera, projects all Gaussians, sorts by camera-space depth, composites, and takes an MSE loss. The run below uses **N = 4000, 1500 steps, Adam at 1e-2**, initialized as a coarse random cloud in a $[-1.5, 1.5]^3$ box with scale 0.08 and identity rotation. The **mean training PSNR is 23.60 dB** (held-out 16.51 dB).



Training view 000: target / render / error (N = 4000, random initialization).

Longer training

Running 4000 steps and evaluating train and held-out PSNR every 250 steps shows training PSNR climbing from 13.2 dB to 25.8 dB while held-out PSNR rises to only about 16.5 dB in the first few hundred steps and then stops improving (it even dips slightly). More steps alone do not close the gap: the cloud fits the training views without learning the scene.

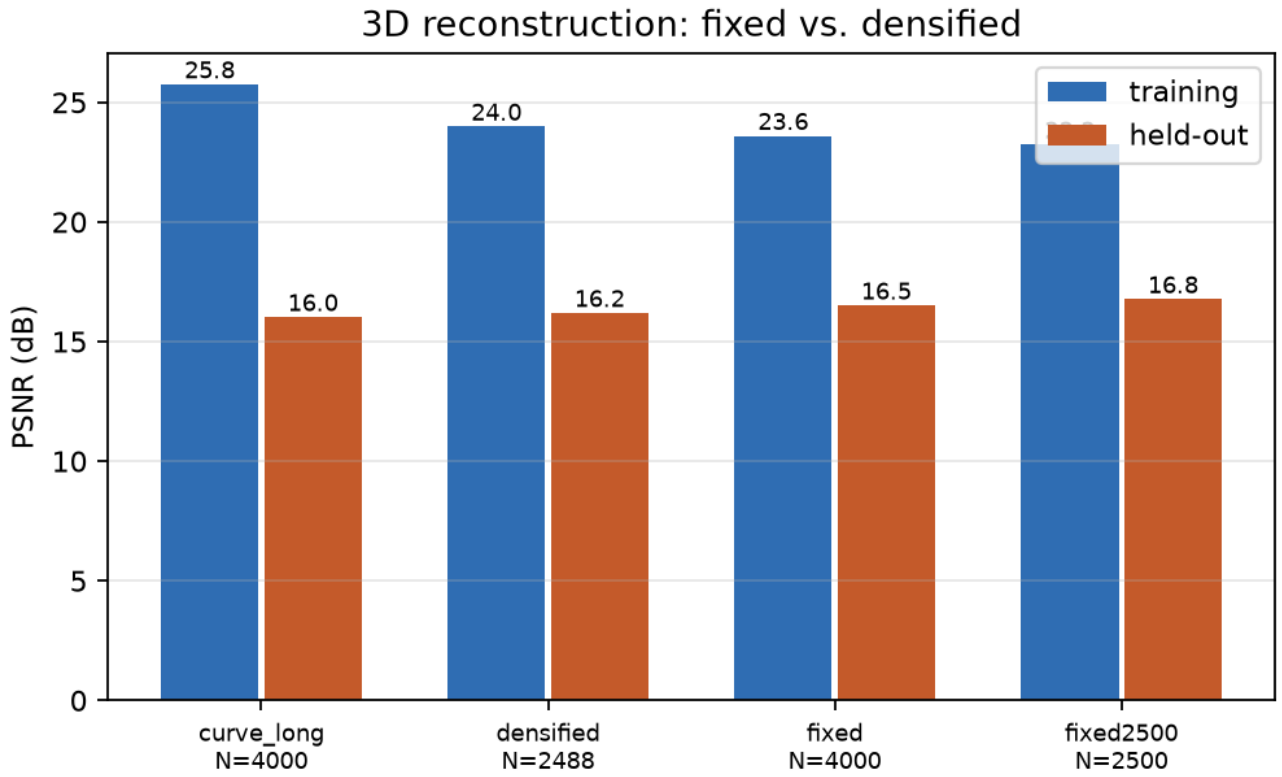


Training and held-out PSNR over 5000 steps (N = 4000).

P8. Densification in 3D

What changed, what stayed. Positions and scales are 3D (`mu` and `log_s` of shape (N, 3)); rotation is a quaternion (N, 4) that clone/split simply copy; the size threshold is in world units instead of pixels; the position-gradient score is accumulated over the randomly sampled training cameras between passes; and pruning the sub-threshold-opacity primitives is what clears floaters. The score, clone-vs-split rule, split factor, budget cap, and Adam-moment bookkeeping are all unchanged.

model	Gaussians	training PSNR	held-out PSNR
fixed	2500	23.24	16.77
densified	2488	23.99	16.20

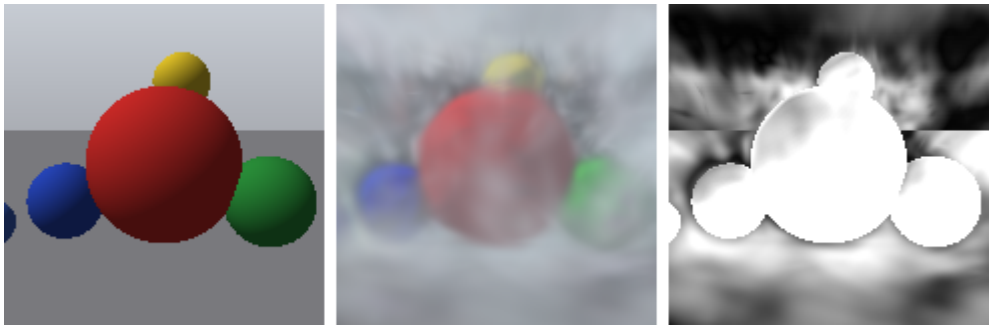


Training and held-out PSNR, fixed vs densified (with the N = 4000 fixed run for reference).

The densified cloud reaches a higher training PSNR but a lower held-out PSNR than the fixed fit at the same count. The extra primitives improve the training views, floaters included, and do not generalize. I report the negative result rather than claim floater reduction: with a per-camera global depth sort and 1500 steps, densification adds training fit without reducing the held-out fog.



Held-out view 000, fixed N = 2500 (16.77 dB held-out).

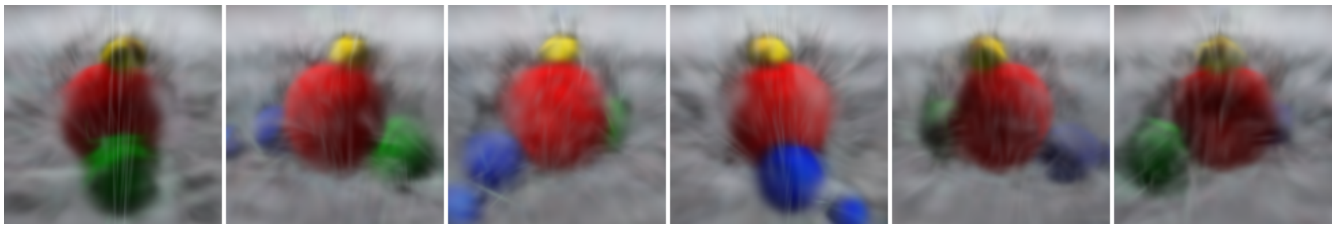


The same held-out view, densified (16.20 dB). Not visibly cleaner; it adds detail to the training views without clearing the held-out fog.

P9. Novel-view evaluation

Held-out vs training. Held-out PSNR is 16.5 dB against 23.6 dB training — a large gap, so a substantial part of the fit is memorization.

Novel view. A 24-frame orbit at 20° elevation (a viewpoint family the optimizer never saw) is in `assets/orbit_fixed.mp4`; six frames:



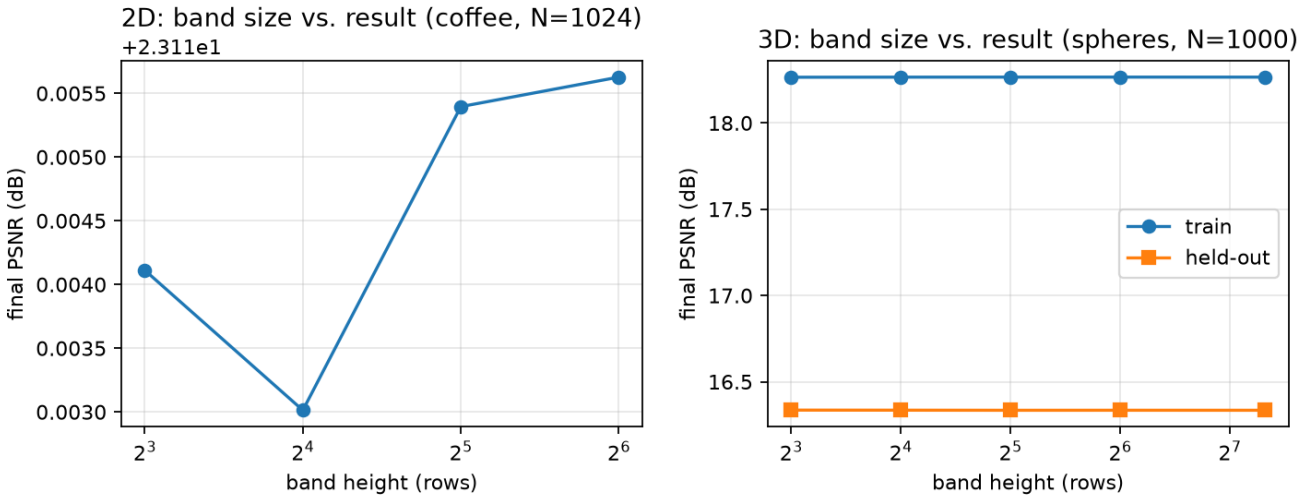
Novel 24-view orbit (20° elevation).

Where it breaks. Sphere silhouettes are soft, thin contacts between spheres merge, and the wall/floor fog drifts with the camera. The causes are the simplifications of this pipeline: a random (not SfM) initialization, a single global depth order rather than a per-pixel one, and view-independent color.

Additional experiment: is the memory batching safe?

The 4 GB GPU cannot hold the full (pixels × Gaussians) weight tensor, so rendering is done in horizontal bands with a local-support cutoff. This must be a pure memory optimization. Retraining the same model under different band heights gives:

band (rows)	2D (coffee, N=1024)	3D train (spheres, N=1000)	3D held-out
8	23.1141	18.2650	16.3355
16	23.1130	18.2655	16.3351
32	23.1154	18.2657	16.3345
64	23.1156	18.2658	16.3347
whole image	out of memory		



Final PSNR against band height — flat to within 0.003 dB.

The spread is floating-point noise (0.003 dB in 2D, 0.001 dB in 3D), and the whole-image 2D configuration does not fit in memory — which is exactly why the bands exist. Disabling the 3.5σ support cutoff entirely changes the held-out PSNR by 0.0002 dB. The batching therefore does not affect the end result.

Extension: Eiffel Tower from internet photos

In progress: 320 Commons photos, COLMAP SfM, RANSAC image selection, and lighting compensation before and during training. Results and the orbit GIF will be added here once the reconstruction finishes.

Conclusions

- The rasterizer and representation are compact and fully differentiable; quality scales smoothly with the number of Gaussians.
- Densification is a clear win in 2D at equal count but did not improve held-out quality in the tested 3D scene.
- The 3D reconstructions are soft because of the global depth order and random initialization; these are the main simplifications versus full 3DGS.
- Batching is a pure memory optimization: identical results across band sizes, and the unbanded variant does not fit in 4 GB.